



网络安全实验

VPN

01

IPSec配置

- Windows IPSec配置
- 抓包分析 IKE SA和IPSec SA建立过程

02

路由器VPN

Cisco Packet Tracer中
VPN配置实验

01

Windows IPSec配置

- Windows IPSec配置
- 抓包分析 IKE SA和IPSec SA建立过程



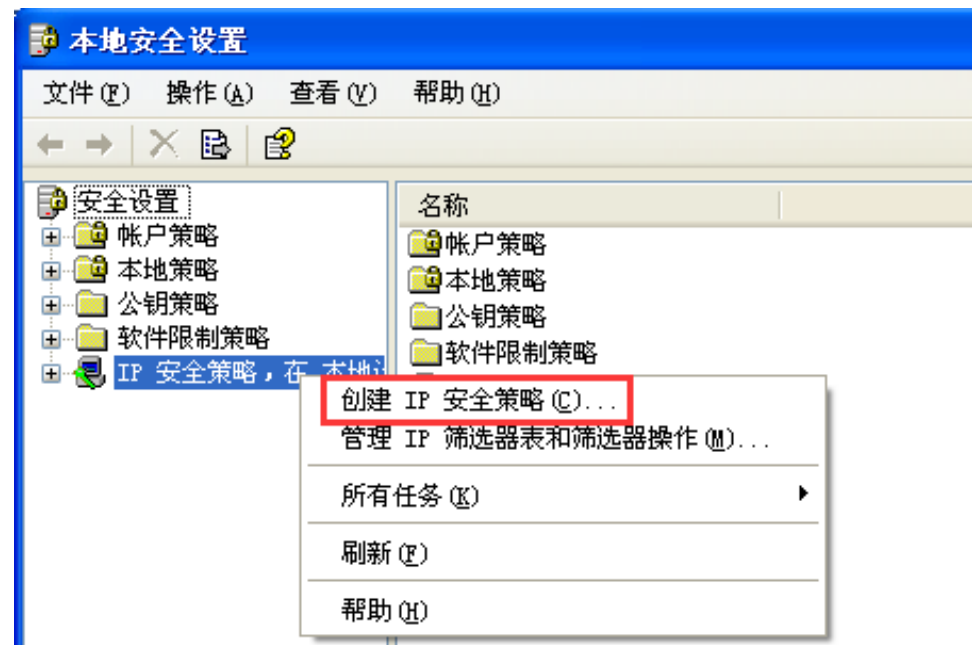
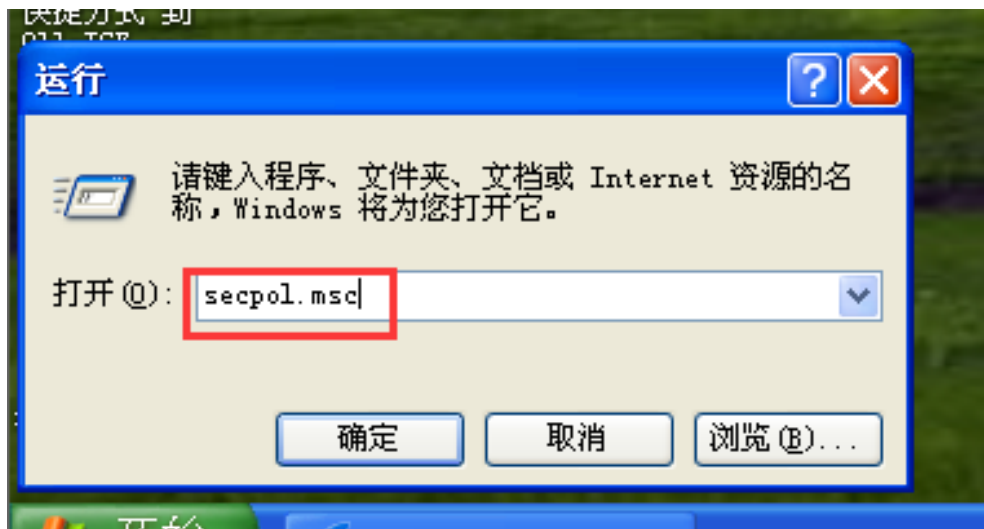
Windows



wireshark

创建安全策略

Win+R输入secpol.msc打开本地安全设置，创建IP安全策略

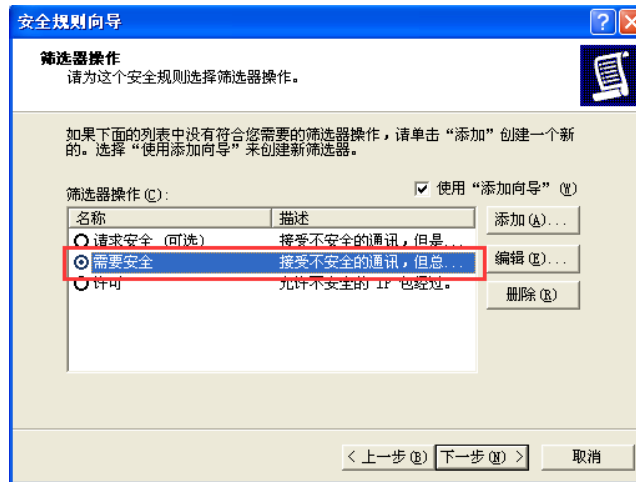
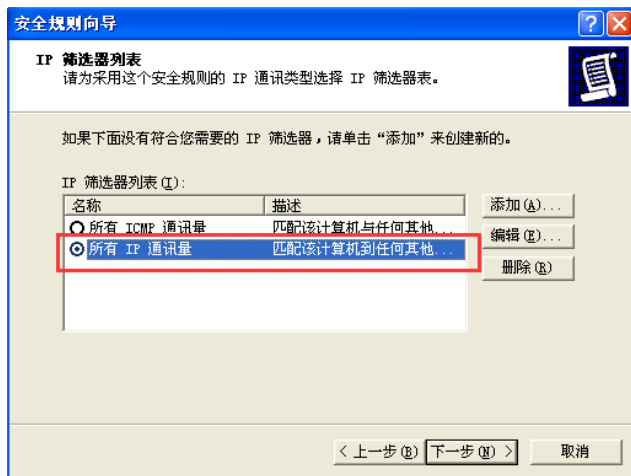




VPN

修改策略属性并指派策略

- 勾选“激活默认相应规则”，设置预共享密钥
- 添加安全规则，不指定隧道终结点，适用于所有网络节点
- 添加IP筛选器列表，添加新筛选器，选择“所有IP通讯量”
- 在筛选器操作界面选择“需要安全”，然后完成修改属性



在另一台Windows主机上执行相似的操作，完成IPSec双方配置



VPN

查看主机网络信息

使用命令ipconfig查看主机IP信息

```
C:\Documents and Settings\Administrator>ipconfig /all

Windows IP Configuration

Host Name . . . . . : winxp-3d8cc5a43
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : localdomain


Ethernet adapter 本地连接:

Connection-specific DNS Suffix . : localdomain
Description . . . . . : VMware Accelerated AMD PCNet Adapter
Physical Address. . . . . : 00-0C-29-C8-16-C1
Dhcp Enabled. . . . . : Yes
Autocconfiguration Enabled. . . . . : Yes
IP Address. . . . . : 192.168.86.141
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.86.2
DHCP Server . . . . . : 192.168.86.254
DNS Servers . . . . . : 192.168.86.2
Primary WINS Server . . . . . : 192.168.86.2
Lease Obtained. . . . . : 2020年11月21日 10:55:51
Lease Expires . . . . . : 2020年11月21日 11:25:51


Ethernet adapter Bluetooth 网络连接:

Media State . . . . . : Media disconnected
Description . . . . . : Bluetooth 设备<个人区域网>
Physical Address. . . . . : 20-16-B9-37-53-97
```

主机通信并抓包

使用主机1去ping主机2，使用WireShark抓包

```
C:\Users\>ping 192.168.86.141

正在 Ping 192.168.86.141 具有 32 字节的数据:
来自 192.168.86.141 的回复: 字节=32 时间=4ms TTL=128
来自 192.168.86.141 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.86.141 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.86.141 的回复: 字节=32 时间<1ms TTL=128

192.168.86.141 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
    往返行程的估计时间(以毫秒为单位):
        最短 = 0ms, 最长 = 4ms, 平均 = 1ms
```

Wireshark 1.10.0 (SVN Rev 49790 from /trunk-1.10)

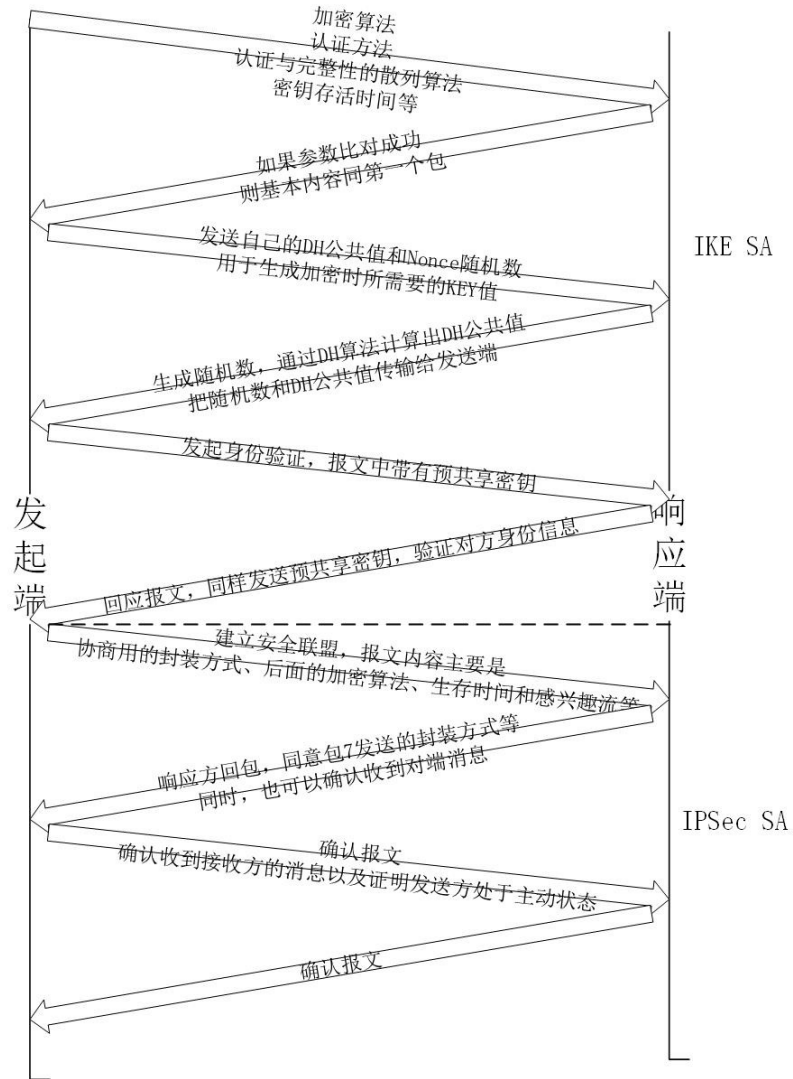
Filter: ip.addr eq 192.168.86.141 and ip.addr eq 192.168.86.136

No.	Time	Source	Destination	Protocol	Length	Info
42	16.6169220	192.168.86.136	192.168.86.141	ISAKMP	294	Identity Protection (Main Mode)
45	16.6177350	192.168.86.141	192.168.86.136	ISAKMP	190	Identity Protection (Main Mode)
46	16.6187820	192.168.86.136	192.168.86.141	ISAKMP	302	Identity Protection (Main Mode)
47	16.6275680	192.168.86.141	192.168.86.136	ISAKMP	274	Identity Protection (Main Mode)
48	16.6284040	192.168.86.136	192.168.86.141	ISAKMP	110	Identity Protection (Main Mode)
49	16.6286930	192.168.86.141	192.168.86.136	ISAKMP	110	Identity Protection (Main Mode)
50	16.6300060	192.168.86.136	192.168.86.141	ISAKMP	214	Quick Mode
51	16.6304100	192.168.86.141	192.168.86.136	ISAKMP	230	Quick Mode
52	16.6334310	192.168.86.136	192.168.86.141	ISAKMP	102	Quick Mode
53	16.6337240	192.168.86.141	192.168.86.136	ISAKMP	126	Quick Mode
54	16.6346130	192.168.86.136	192.168.86.141	ESP	110	ESP (SPI=0xf2baabc1)
55	16.6346680	192.168.86.141	192.168.86.136	ESP	110	ESP (SPI=0x4a9027b0)
62	17.6247470	192.168.86.136	192.168.86.141	ESP	110	ESP (SPI=0xf2baabc1)
63	17.6248030	192.168.86.141	192.168.86.136	ESP	110	ESP (SPI=0x4a9027b0)
66	18.6410260	192.168.86.136	192.168.86.141	ESP	110	ESP (SPI=0xf2baabc1)
67	18.6411010	192.168.86.141	192.168.86.136	ESP	110	ESP (SPI=0x4a9027b0)
69	19.6553340	192.168.86.136	192.168.86.141	ESP	110	ESP (SPI=0xf2baabc1)
70	19.6553910	192.168.86.141	192.168.86.136	ESP	110	ESP (SPI=0x4a9027b0)



VPN

分析IKE SA和IPSec SA建立过程



第一个数据包

```
Wireshark - 分组 50 - Ethernet II
> Frame 50: 294 bytes on wire (2352 bits), 294 bytes captured (2352 bits) on interface \Device\NPF_{9AA21F26-638E-41D6-B614-4...}
> Ethernet II, Src: VMWare_26:00:08 (00:0c:29:26:00:08), Dst: VMWare_c8:16:c1 (00:0c:29:c8:16:c1)
> Internet Protocol Version 4, Src: 192.168.86.136, Dst: 192.168.86.141
> User Datagram Protocol, Src Port: 500, Dst Port: 500
> Internet Security Association and Key Management Protocol
  Initiator SPI: 627bdf3554ad077
  Responder SPI: 0000000000000000
  Next payload: Security Association (1)
  Version: 1.0
  Exchange Type: Identity Protection (Main Mode) (2)
  Flags: 0x00
  Message ID: 0x00000000
  Length: 252
  Payload: Security Association (1)
    Payload: Vendor ID (13) : Unknown Vendor ID
    Payload: Vendor ID (13) : MS MT5 ISAKMPKEY
    Payload: Vendor ID (13) : RFC 3947 Negotiation of NAT-Traversal in the IKE
    Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-02/n
    Payload: Vendor ID (13) : Cisco Fragmentation
    Payload: Vendor ID (13) : MS-Negotiation Discovery Capable
    Payload: Vendor ID (13) : Microsoft Vid-Initial-Contact
    Payload: Vendor ID (13) : IKE Gex Version 1
```

- ▼ Payload: Proposal (2) # 1
 - Next payload: NONE / No Next Payload (0)
 - Reserved: 00
 - Payload length: 44
 - Proposal number: 1
 - Protocol ID: ISAKMP (1)
 - SPI Size: 0
 - Proposal transforms: 1
- ▼ Payload: Transform (3) # 1
 - Next payload: NONE / No Next Payload (0)
 - Reserved: 00
 - Payload length: 36
 - Transform number: 1
 - Transform ID: KEY_IKE (1)
 - Reserved: 0000
 - Encryption-Algorithm: 3DES-CBC
 - Hash-Algorithm: SHA
 - Group-Description: Alternate 1024-bit MODP group
 - Authentication-Method: Pre-shared key
 - Life-Type: Seconds
 - Life-Duration: 28800
- ▼ Payload: Vendor ID (13) : Unknown Vendor ID

第七个数据包

```
Wireshark - 分组 58 - Ethernet II
> Frame 58: 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface \Device\NPF_{9AA21F26-638E-41D6-B614-4...}
> Ethernet II, Src: VMWare_26:00:08 (00:0c:29:26:00:08), Dst: VMWare_c8:16:c1 (00:0c:29:c8:16:c1)
> Internet Protocol Version 4, Src: 192.168.86.136, Dst: 192.168.86.141
> User Datagram Protocol, Src Port: 500, Dst Port: 500
> Internet Security Association and Key Management Protocol
  Initiator SPI: 627bdf3554ad077
  Responder SPI: 538309c00167714c
  Next payload: Hash (8)
  Version: 1.0
  Exchange type: Quick Mode (32)
  Flags: 0x01
    ... .1 = Encryption: Encrypted
    ... .0 = Commit: No commit
    ... .0 = Authentication: No authentication
  Message ID: 0x00000001
  Length: 172
  Encrypted Data (144 bytes)
```

第十个数据包

```
Wireshark - 分组 61 - Ethernet II
> Frame 61: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface \Device\NPF_{9AA21F26-638E-41D6-B614-4...}
> Ethernet II, Src: VMWare_26:00:08 (00:0c:29:26:00:08), Dst: VMWare_c8:16:c1 (00:0c:29:c8:16:c1)
> Internet Protocol Version 4, Src: 192.168.86.141, Dst: 192.168.86.136
> User Datagram Protocol, Src Port: 500, Dst Port: 500
> Internet Security Association and Key Management Protocol
  Initiator SPI: 627bdf3554ad077
  Responder SPI: 538309c00167714c
  Next payload: Hash (8)
  Version: 1.0
  Exchange type: Quick Mode (32)
  Flags: 0x03
    ... .1 = Encryption: Encrypted
    ... .0 = Commit: Commit
    ... .0 = Authentication: No authentication
  Message ID: 0x00000001
  Length: 84
  Encrypted Data (56 bytes)
```



02

路由器VPN

Cisco Packet Tracer中VPN配置实验



Cisco Packet Tracer配置VPN

安装PacketTracer

安装后使用游客身份登录



Are you a new user?

Sign up for Cisco Networking Academy and get free access to Packet Tracer and online course.

[Sign Up](#)

Continue as a Guest. Guest users can save files up to 3 times.

[Continue as a Guest](#)

安装完成后以游客身份使用



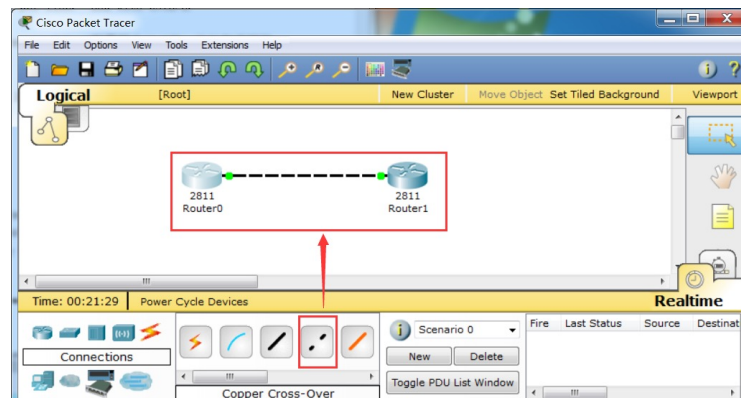
VPN

搭建网络环境

- 初始化路由器
- 配置PC和交换机
- 配置路由器使网络互通

```
Router0
Physical Config CLI
IOS Command Line Interface
Continue with configuration dialog? [yes/no]: n
Press RETURN to get started!

Router>enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
$LINK-S-CHANGED: Interface FastEthernet0/0, changed state to up
end
Router#
$SYS-S-CONFIG_1: Configured from console by console
configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet 0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
$LINK-S-CHANGED: Interface FastEthernet0/1, changed state to up
end
Router#
$SYS-S-CONFIG_1: Configured from console by console
```



PC0

Physical Config Desktop

IP Configuration

☐ DHCP
☒ Static

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

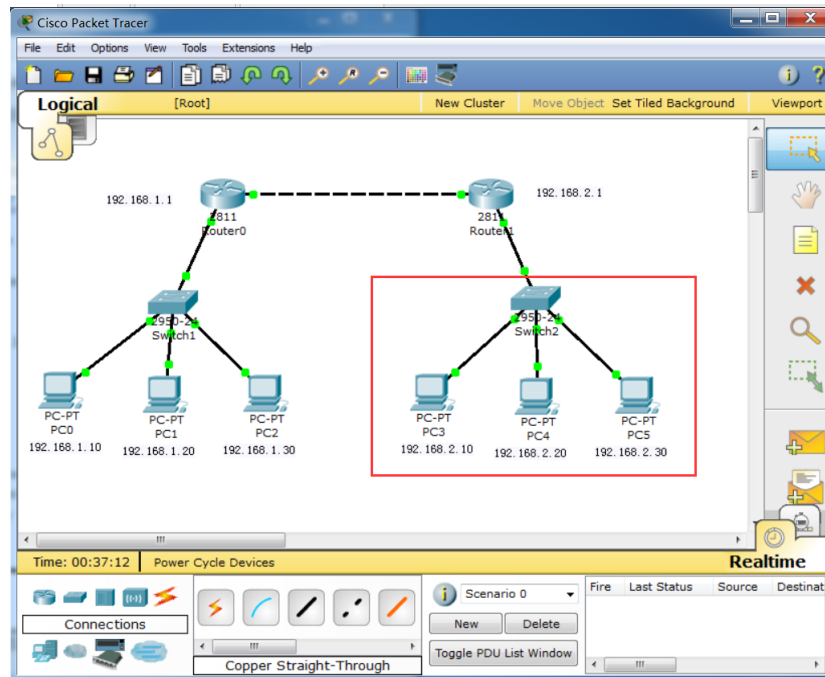
DNS Server:



VPN

测试网络连通性

- PC0能ping通router0的内外网地址
- PC0能ping通router1的内外网地址
- PC0能ping通router1连接的主机地址



Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=110ms TTL=255
Reply from 192.168.1.1: bytes=32 time=46ms TTL=255
Reply from 192.168.1.1: bytes=32 time=63ms TTL=255
Reply from 192.168.1.1: bytes=32 time=62ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 46ms, Maximum = 110ms, Average = 70ms
```

```
PC>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=93ms TTL=254
Reply from 10.0.0.2: bytes=32 time=93ms TTL=254
Reply from 10.0.0.2: bytes=32 time=94ms TTL=254
Reply from 10.0.0.2: bytes=32 time=93ms TTL=254

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 93ms, Maximum = 94ms, Average = 93ms
```

```
PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=156ms TTL=126
Reply from 192.168.2.10: bytes=32 time=140ms TTL=126
Reply from 192.168.2.10: bytes=32 time=141ms TTL=126
Reply from 192.168.2.10: bytes=32 time=140ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 140ms, Maximum = 156ms, Average = 144ms
```



VPN

配置IPSec VPN

- 配置router0和router1

测试IPSec VPN

- 测试VPN连通性
- 验证数据通过IPSec加密传输

```
Router0
Physical Config CLI
IOS Command Line Interface

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#hash md5
Router(config-isakmp)#encryption des
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key bupt address 10.0.0.2
Router(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router(config)#crypto ipsec transform-set mine esp-des esp-md5-hmac
Router(config)#crypto map mymap 101 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#match address 110
Router(config-crypto-map)#set peer 10.0.0.2
Router(config-crypto-map)#set pfs group2
Router(config-crypto-map)#set transform-set mine
Router(config-crypto-map)#set security-association lifetime seconds 86400
Router(config-crypto-map)#interface fastEthernet 0/0
Router(config-if)#crypto map mymap
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: DES - Data Encryption Standard (56 bit keys).
  hash algorithm:      Message Digest 5
  authentication method: Pre-Shared Key
```

定义IKE (密钥交换策略)

定义数据的加密方式和认证方式

配置IPSec

将map映射到公网端口

PDU Information at Device: Router0

At Device: Router0
Source: PC0
Destination: 192.168.2.10

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.10 ICMP Message Type: 8	Layer 3: IP Header Src. IP: 10.0.0.1, Dest. IP: 10.0.0.2 ICMP Message Type: 8
Layer 2: Ethernet II Header 00E0.8F32.7B34 >> 00D0.D314.1A02	Layer 2: Ethernet II Header 00D0.D314.1A01 >> 0030.F270.C801
Layer 1: Port FastEthernet0/1	Layer 1: Port(s): FastEthernet0/0

1. The routing table finds a routing entry to the destination IP address.
2. The destination routing entry is connected static route. The router sets destination as the next-hop.
3. The router decrements the TTL on the packet.
4. The traffic is interesting traffic and needs to be encrypted and encapsulated in IPsec PDUs.
5. The packet is getting encrypted and encapsulated in IPsec PDUs.
6. ESP encrypts the received packet.
7. The router encapsulates the data into an IP packet.
8. The router looks up the destination IP address in the routing table.
9. The routing table finds a routing entry to the destination IP address.
10. The destination network is directly connected. The router sets destination as the next-hop.

Challenge Me << Previous Layer Next Layer >>



谢谢！